Rigid Folding of Periodic Triangulated Origami Tessellations

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Abstract. In this paper, we investigate the continuous rigid folding motion of an origami tessellation structure when their pattern is infinitely tessellating a plane with periodic symmetry. We analytically and numerically describe the kinematics of triangular origami tessellation by assuming that the fold configuration, as well as the pattern, is periodic. Owing to the geometric constraints, the global forms of these tessellations are always restricted to cylindrical configurations unlike the folding of a finite open origami tessellation, which allows a double curved surface. In such triangulated patterns, the number of fold angles representing the configuration of the fundamental figure is equal to the number of equations from rigid origami constraints. However, this forms a two degrees of freedom mechanism instead of being statically determinate. One of the degrees of freedom represents the folding-unfolding motion, and the other represents the change in the orientation of the rolling axis.

1. Introduction

Repetitive origami tessellations, such as Miura, Yoshimura, Resch, and Waterbomb tessellations, are attracting the attention of engineers and designers; their applications include reconfigurable mechanisms in both small and large scales [**2, 7**], as well as lightweight material such as the fold core of a sandwich panel [**3**]. In such applications, we use hard materials such as thick panels and metal sheets instead of soft paper; the folding motion of such materials can be modeled as that of rigid origami, in which the bending of material or continuous traveling of creases is forbidden. In order to materialize designs using origami tessellations, it is important to study the continuous rigid-folding of a planar pattern into the 3D tessellated form. For example, a smaller portion of the tessellation can be naturally folded into a double curved synclastic surface, with a less folded part in the center and a more folded part around the boundary (Figure 1, Left). However, if we increase the number of repeating modules, we can observe that the heterogeneity of the folding will block the rigid folding process, as can be seen in Figure 1, Right. This poses a question: is rigid folding of the entire tessellation a universal behavior, scalable to different industrial purposes, or just a boundary effect that is limited by the size of the pattern?

In this paper, we investigate the continuous rigid folding motion of triangulated origami tessellations when their crease patterns are periodically tessellating an infinite plane. Presently, we also assume the repeating symmetry of the folding angles,

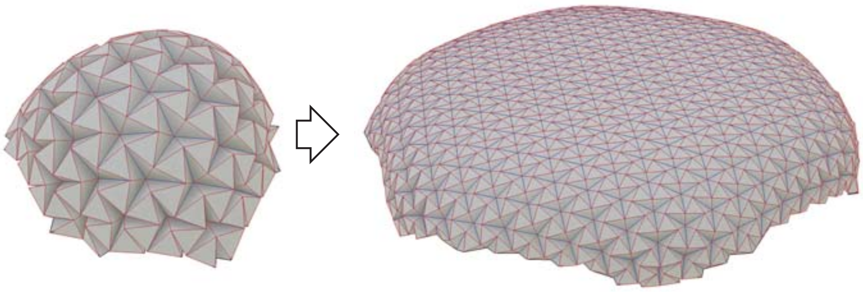

FIGURE 1. Left: Small portion of Resch's triangular pattern forming a synclastic surface. Right: Larger portion of the pattern. The folding motion is obstructed when the center part remains almost unfolded.

so that we can avoid the heterogeneity of the folding angles, as the heterogeneity potentially blocks the continuous folding paths from a flat state to the completely folded state. An initial and naive approach toward understanding the flexibility of a rigid origami system is to compare the number of variables and constraints; the variable fold angles are constrained by the identity of rotation around interior vertices [**1, 4**]. Therefore, an origami pattern is estimated to be rigid foldable if $E - 3V > 0$, where E and V denote the numbers of foldlines and interior vertices, respectively, while every periodic triangular patten follows $E = 3V$. This suggests the non-rigid foldability of generic triangular pattern. However, in a numeric simulation, it is observed that a generic triangular tessellation forms a 2-degrees of freedom (DOF) mechanism. We will identify the degeneracy of the constraints originating in the periodic symmetry of the tessellations, and demonstrate different behaviors for different patterns. In addition, we show that the folded form of repetitive tessellations are normally constrained to cylindrical surfaces, as opposed to the behavior of a finite portion of the tessellation, which can be folded into a double curved surface.

2. Rigid Origami Basics

Since each panel is planar in rigid origami, we assume that creases are straight. In addition, we allow the surfaces to globally intersect, as most of the patterns intersect themselves if the surface is expanded to infinity.

DEFINITION 2.1. A *crease pattern* C on a planar region P is a planar straight line graph embedding that partitions P. P denotes a piece of paper.

DEFINITION 2.2. A *rigid folding* $f : \mathbb{R}^2 \to \mathbb{R}^3$ (allowing a collision) by crease pattern C is an intrinsically isometric transformation of a piece of paper P in $3D$, such that each planar region partitioned by C is congruently mapped, i.e., kept planar. The *folded form* of C is the image $f(P)$. This forms an orientable polyhedral 2-manifold, where its set of non-differentiable points lie on the image $f(C)$; for each segment e of C, termed a *crease*, we can assign the *fold angles* ρ_e in $(-\pi, \pi)$ by measuring the exterior dihedral angle at the folded form of the crease.

Here it is possible that the crease is mapped onto a planar region, in which case the fold angle is 0. We call the edges *mountains* or *valleys*, when their fold angles are either negative or positive, respectively.

We use the fold angle based representation of rigid origami.

DEFINITION 2.3. A *single crease fold* $\mathbf{R}_e$, by crease e, in direction $\mathbf{t}$ (not parallel to e) is the rotation about an axis along e orienting to the right side of $\mathbf{t}$, by a folding angle ρ_e of e. For a given oriented curve γ on a plane, not passing though vertices and not tangent to creases, a *relative rigid folding* $\mathbf{F}(\gamma)$ by a crease pattern C along γ is the rigid transformation formed by the product of single crease folds by intersecting creases of C, in tangent directions of γ at the intersections, where the product is calculated in the order along γ. The flipped path of γ is denoted by $\overleftarrow{\gamma}$, and $\mathbf{F}(\overleftarrow{\gamma}) = \mathbf{F}^{-1}(\gamma)$.

A relative rigid folding $\mathbf{F}(\gamma)$ can be represented by a homogeneous matrix, using single crease folds $\mathbf{R}_i$ of the i-th edge along γ:

$$\mathbf{F}(\gamma) = \mathbf{R}_1\mathbf{R}_2 \cdots \mathbf{R}_n, \tag{2.1}$$

where $\mathbf{R}_i$ can be decomposed as

$$\begin{aligned}\mathbf{R}_i &= \mathbf{Y}_i\mathbf{P}_i\mathbf{Y}_i^{-1} \\ &= \begin{bmatrix} c_i & -s_i & 0 & -rs_i \\ s_i & c_i & 0 & rc_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\rho_i & -\sin\rho_i & 0 \\ 0 & \sin\rho_i & \cos\rho_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_i & s_i & 0 & 0 \\ -s_i & c_i & 0 & -r \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\end{aligned}$$

where the rotation axis passing along crease i is represented by its normalized orientation vector $\mathbf{l}_i = (c_i, s_i) = (\cos\theta_1, \sin\theta_i)$ and its shortest distance from the origin r. The sign of θ is determined such that $\mathbf{l}_i$ orients to the right side with respect to the tangent vector of γ at the intersection point. Subsequently, rigid foldaiblity, allowing intersection, is given as follows.

THEOREM 2.4. *A rigid folding (allowing intersection) of a planar region by a crease pattern with given fold angles exists if and only if for any simple closed curve on a plane, not passing though vertices and not tangent to creases, the relative folding along the curve is the identity.*

This can be written as

$$\mathbf{F}(\gamma) = \mathbf{R}_0\mathbf{R}_1 \cdots \mathbf{R}_{n-1} = \mathbf{I}. \tag{2.2}$$

This is implied in the works of Kawasaki [**1**] and belcastro and Hull [**4**]. However, we give a separate proof in the Appendix. Theorem 2.4 can be reduced in the case where the paper is homeomorphic to a disk, as follows.

DEFINITION 2.5. The *single-vertex compatibility* of an interior vertex is the property that the relative rigid folding along a small circle centered at the vertex is equal to the identity, where small means that the circle intersects only with creases that are incident to the vertex.

THEOREM 2.6. *A rigid folding (allowing intersection) of a planar disk region by a crease pattern with given fold angles exists if and only if single-vertex compatibility is satisfied for every interior vertex.*

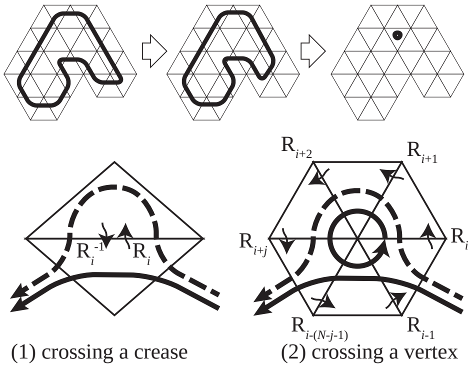

FIGURE 2. Contraction of a curve.

PROOF. The necessity follows from Theorem 2.4. In order to show sufficiency, we consider any simple closed loop γ and show that it satisfies $\mathbf{F}(\gamma) = \mathbf{I}$, from the conditions around each vertex. Consider a continuous contraction of γ to a point and its corresponding events, i.e., the change in the sequence of creases $e_0, e_1, \ldots, e_{n-1}$. There are two possible types of such events; (1) a curve crosses a crease, and (2) a curve crosses an interior vertex (Figure 2). In case (1), for the curves γ and γ' before and after crossing, $\mathbf{F}(\gamma)$ and $\mathbf{F}(\gamma')$ are equal, since

$$\begin{aligned} \mathbf{F}(\gamma) &= \mathbf{R}_A \left(\mathbf{R}_i \mathbf{R}_i^{-1}\right) \mathbf{R}_B \\ \mathbf{F}(\gamma') &= \mathbf{R}_A \mathbf{R}_B, \end{aligned}$$

where $\mathbf{R}_A$ and $\mathbf{R}_B$ represent the product of the rotations that are shared by both curves. Similarly, in the case of (2),

$$\begin{aligned} \mathbf{F}(\gamma) &= \mathbf{R}_A \left(\mathbf{R}_i \mathbf{R}_{i+1} \cdots \mathbf{R}_{i+j}\right) \mathbf{R}_B \\ \mathbf{F}(\gamma') &= \mathbf{R}_A \left(\mathbf{R}_{i-1}^{-1} \mathbf{R}_{i-2}^{-1} \cdots \mathbf{R}_{i-(N-j-1)}^{-1}\right) \mathbf{R}_B. \end{aligned}$$

Here, the single-vertex compatibility of the interior vertex crossed by γ gives $\mathbf{R}_{i-1}^{-1}$ $\mathbf{R}_{i-2}^{-1} \cdots \mathbf{R}_{i-(N-j-1)}^{-1} = \mathbf{R}_i \mathbf{R}_{i+1} \cdots \mathbf{R}_{i+j}$. Therefore, $\mathbf{F}(\gamma) = \mathbf{F}(\gamma')$. Since γ is contracted to a point, $\mathbf{F}(\gamma) = \mathbf{I}$. □

In the case of a disk with a hole, a necessary and sufficient condition is as follows.

THEOREM 2.7. *A rigid folding (allowing intersection) of a planar disk with a hole (the two boundaries do not share a point) by a crease pattern with given fold angles exists if and only if single-vertex compatibility is satisfied for every interior vertex, and there exists a closed curve around the hole such that the relative folding along the curve is the identity.*

PROOF. The necessity follows from Theorem 2.4. In order to show sufficiency, consider the contraction process as in the case of a disk. A closed loop that can be contracted to a point is the same as in the case of a disk.If a closed loop cannot be contracted to a point, then it is a loop around a hole. Any such loops can be continuously transformed to each other through the sequence of events (1) and (2) in the previous proof, and thus the relative folding is equal to the identity. □

Note that in the case of the disk, even though it is written as a 4×4 matrix equation, the single-vertex compatibility of each interior vertex essentially yields three equality constraints, as the creases pass through a common vertex, and the translation term disappears. These constraints form a system of variables of fold angles $\rho_1, \ldots, \rho_E$, with V rotational constraints, i.e., $3V$ equations, imposed on them, where E and V are the numbers of interior edges and interior vertices, respectively. Therefore, rigid foldability and the number of DOF can be first estimated through $M := E - 3V$. If there is no singularity in the system, $M > 0$ represents the rigid foldability and the nubmer of DOF is M.

2.1. Infinitesimal Folding. How does the rigid origami transform? One way to look at it (and to simulate rigid origami) is to consider the infinitesimal folding motion. Consider a loop, passing through creases $0, 1, \ldots, i, \ldots, n-1$ in this order, and the folding of the crease pattern such that the facet before crease 0 does not move. In such a folded state, let $\mathbf{L}_i = (L_i^x, L_i^y, L_i^z)^\mathsf{T}$ denote the orientation of crease i, and $\mathbf{O}_i$ denote a (any) point on the crease. Then differentiating Equation 2.1 gives

$$\sum_i \frac{\partial \mathbf{F}(\gamma)}{\partial \rho_i} d\rho_i = \mathbf{0},$$

where

$$\begin{aligned} \frac{\partial \mathbf{F}(\gamma)}{\partial \rho_i} &= \mathbf{R}_0 \mathbf{R}_1 \cdots \frac{\partial \mathbf{R}_i}{\partial \rho_i} \cdots \mathbf{R}_{n-1} \\ &= \left[\begin{array}{ccc|c} 0 & -L_i^z & L_i^y & \\ L_i^z & 0 & -L_i^x & \mathbf{L}_i \times \mathbf{O}_i \\ -L_i^y & L_i^x & 0 & \\ \hline 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$$

This is equivalent to

$$\left\{ \begin{array}{lcl} \sum_i \mathbf{L}_i d\rho_i & = & \mathbf{0} \\ \sum_i \mathbf{L}_i d\rho_i \times \mathbf{O}_i & = & \mathbf{0} \end{array} \right. . \tag{2.3}$$

This means that the infinitesimal motion of rigid origami with a single vertex or multiple vertices can be described as the equilibrium of a point or a rigid body, respectively, when we substitute axial force for angular velocity [1]. As rigid-foldability implies infinitesimal rigid-foldability, the dimension of the solution space of the linear system in Equation 4 is greater than or equal to the number of DOF in the actual rigid-folding system.

3. Periodic Origami Tessellations

In the following, we will detail the property of the folded form of periodic origami tessellations (if it exists), as well as the proof of its actual existence, and the number of DOF of the mechanism.

Definition 3.1. A *periodic crease pattern* C is a crease pattern on a full plane that partitions the plane into periodic polygonal tiles, such that the symmetry group G is wallpaper group $p2$.

[1]This interesting duality is also straightforwardly described in terms of screw theory in kinematics

DEFINITION 3.2. A *symmetric rigid folding* by a periodic crease pattern C is a rigid folding by C, such that any pair of creases e and f of C that map onto each other by transformation in G share the same fold angle $\rho_e = \rho_f$.

DEFINITION 3.3. A *fundamental region* D of a periodic crease pattern C is a planar disk region that tessellates a plane by the symmetry group G of C, such that its boundary does not pass through the vertex of C. A *fundamental crease pattern* C_D of D is a subset of C within D.

DEFINITION 3.4. For a periodic crease pattern C and its generating translations $\mathbf{T}_1$ and $\mathbf{T}_2$, define two curves γ_1 and γ_2 sharing the starting point $\mathbf{x}$ and ending at $\mathbf{T}_1\mathbf{x}$ and $\mathbf{T}_2\mathbf{x}$, respectively. We call γ_1 and γ_2 *generating paths* if a joined path γ_1, $\mathbf{T}_1\gamma_2$, $\mathbf{T}_2\overleftarrow{\gamma}_1$, and $\overleftarrow{\gamma}_2$ forms the boundary of a fundamental region.

3.1. Cylindrical Surface. If there exists a periodic rigid folding, then in a generic case, the folded form approximately follow a cylindrical surface.

LEMMA 3.5. *Consider a pair of generating paths γ_1 and γ_2, the relative foldings $\mathbf{F}(\gamma_1)$ and $\mathbf{F}(\gamma_2)$ along the paths, and their corresponding translations $\mathbf{T}_1$ and $\mathbf{T}_2$, respectively. Then the relative folding along the boundary of the corresponding fundamental figure is equal to the identity if and only if $\mathbf{F}(\gamma_1)\mathbf{T}_1$ and $\mathbf{F}(\gamma_2)\mathbf{T}_2$ are commutative.*

PROOF. Consider the four corner facets of the fundamental figure A, $\mathbf{T}_1A$, $\mathbf{T}_1\mathbf{T}_2A$, $\mathbf{T}_2A$. Without loss of generality, the folding f is assumed not to transform facet A, i.e., $A = f(A)$. Since $\mathbf{T}_2$ translates the set of creases intersecting γ_1 onto the creases intersecting $\mathbf{T}_2\gamma_1$,

$$\mathbf{F}(\mathbf{T}_2\gamma_1) = \mathbf{T}_2\mathbf{F}(\gamma_1)\mathbf{T}_2^{-1}.$$

In a similar way,

$$\mathbf{F}(\mathbf{T}_1\gamma_2) = \mathbf{T}_1\mathbf{F}(\gamma_2)\mathbf{T}_1^{-1}.$$

Then,

$$\begin{aligned}
\mathbf{F}(\gamma_1, \mathbf{T}_1\gamma_2, \mathbf{T}_2\overleftarrow{\gamma}_1, \overleftarrow{\gamma}_2) &= \mathbf{F}(\gamma_1)\mathbf{F}(\mathbf{T}_1\gamma_2)\mathbf{F}(\mathbf{T}_2\gamma_1)^{-1}\mathbf{F}(\gamma_2)^{-1} \\
&= \mathbf{F}(\gamma_1)(\mathbf{T}_1\mathbf{F}(\gamma_2)\mathbf{T}_1^{-1})\left(\mathbf{T}_2\mathbf{F}(\gamma_1)\mathbf{T}_2^{-1}\right)^{-1}\mathbf{F}(\gamma_2)^{-1} \\
&= \left(\mathbf{F}(\gamma_1)\mathbf{T}_1\right)\left(\mathbf{F}(\gamma_2)\mathbf{T}_2\right)\left(\mathbf{F}(\gamma_1)\mathbf{T}_1\right)^{-1}\left(\mathbf{F}(\gamma_2)\mathbf{T}_2\right)^{-1}. \qquad (3.1)
\end{aligned}$$

Here, we used the commutativity of translations $\mathbf{T}_1^{-1}\mathbf{T}_2 = \mathbf{T}_2\mathbf{T}_1^{-1}$. Therefore, the left-hand side is equal to the identity if and only if $\mathbf{F}(\gamma_1)\mathbf{T}_1$ and $\mathbf{F}(\gamma_2)\mathbf{T}_2$ are commutative. □

THEOREM 3.6. *The folded form of a symmetric rigid folding by a periodic crease pattern is periodic, and its symmetry group is generated by two commutative rigid body displacements.*

PROOF. Consider a periodic crease pattern C, whose symmetry group G is generated by two linearly independent translations $\mathbf{T}_1$, $\mathbf{T}_2$. For an arbitrary facet A and its translated copy $\mathbf{T}_1A$, we consider their folded form $f(A)$ and $f(\mathbf{T}_1A)$ by rigid folding f (Figure 3). Define congruent transformations $\mathbf{S}_1$ and $\mathbf{S}_2$ as $f(\mathbf{T}_1A) = \mathbf{S}_1f(A)$ and $f(\mathbf{T}_2A) = \mathbf{S}_2f(A)$. Without loss of generality, we assume that $f = f(A)$. Then $\mathbf{S}_1 = \mathbf{F}(\gamma_1)\mathbf{T}_1$ and $\mathbf{S}_2 = \mathbf{F}(\gamma_2)\mathbf{T}_2$, where we follow the notations in Lemma 3.5. From Lemma 3.5, $\mathbf{S}_1$ and $\mathbf{S}_2$ are commutative. Because of the periodic crease pattern, the configuration of the creases and fold angles

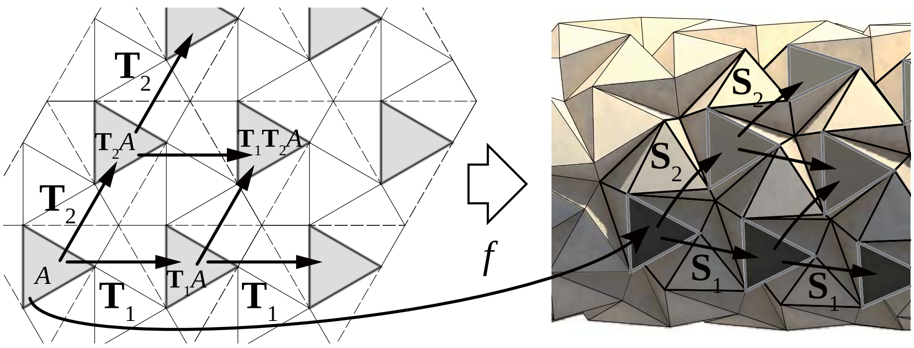

FIGURE 3. Symmetric rigid folding of a periodic crease pattern forms a periodic folded form.

around A and $\mathbf{T}_1A$ are the same. As a result of this property, any disk portion of the folded form around $f(A)$ is congruently mapped to that of around $f(\mathbf{T}_1A)$ by $\mathbf{S}_1$. Therefore, for any facet B, $f(\mathbf{T}_1B) = \mathbf{S}_1f(B)$ is satisfied, and thus $f(\mathbf{T}_1^nB) = \mathbf{S}_1f(\mathbf{T}_1^{n-1}B) = \mathbf{S}_1^nf(B)$. In a similar way, $f(\mathbf{T}_2^nB) = \mathbf{S}_2^nf(B)$. For any copy $B_{m,n} = \mathbf{T}_1^m\mathbf{T}_2^nB$ $(n, m \in \mathbb{Z})$, we can obtain $f(B_{m,n}) = \mathbf{S}_1^m\mathbf{S}_2^nf(B)$. Therefore, the folded form is periodic, and its symmetry group is generated by two commutative rigid transformations $\mathbf{S}_1$ and $\mathbf{S}_2$. □

A rigid transformation is represented as a screw motion, including singular cases of rotation, translation, and identity. A screw motion is represented by 6 parameters, i.e., axis orientation (2-DOF) and positions (2-DOF), rotation angle (1-DOF), and translation along the axis (1-DOF). Two rigid transformations are commutative in the following cases:

(1) two screws/rotations sharing a common axis
(2) a screw/rotation and a translation parallel to the axis of the other
(3) two translations
(4) an arbitrary motion and an identity transformation
(5) two 180° rotations whose axes cross perpendicularly. This cannot actually happen because the developability is violated[2].

The unfolded state follows case (3), while a generic case follows either case (1) or (2), resulting in an approximately cylindrical surface whose axis is located on the shared axis (Figure 4). An interesting example of case (4) is given by a torus origami, as shown in Fig 5.

3.2. Rigid Foldability. Does a periodic origami tessellation actually rigidly fold? Here, we assume that every panel is triangulated to maximize the flexibility. Let us count the number of variables and constraints. Because of the translational symmetry in two directions, the Euler characteristic of the fundamental crease pattern is equal to that of a torus, i.e., $V - E + F = 0$, where F is the number of facets. Because facets are triangulated, we have that $F = \frac{2}{3}E$, which leads to $M = E - 3V = 0$. Therefore, we might initially estimate that generic periodic

[2]Consider a gauss map along a closed loop corresponding to a sequence of $\mathbf{S}_1$, $\mathbf{S}_2$, $\mathbf{S}_1^{-1}$, $\mathbf{S}_2^{-1}$. Then the path turns by 90° at each corner in the same orientation, leading to integral curvature of $360° \neq 0$.

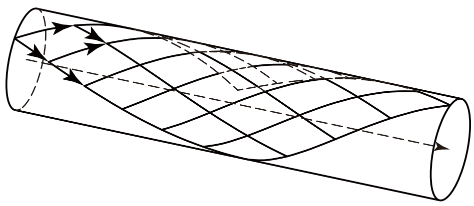

FIGURE 4. Two motions sharing a common screw axis approximately form a cylindrical surface.

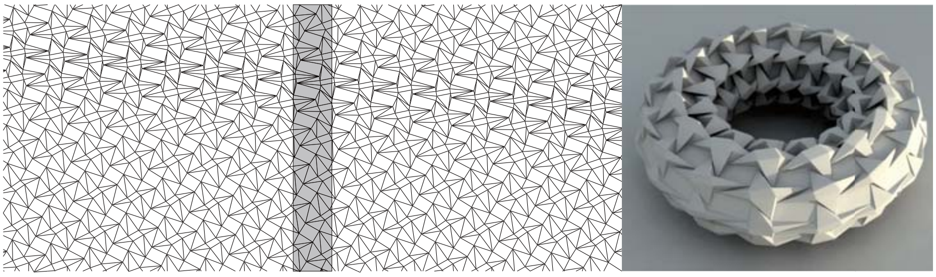

FIGURE 5. A torus origami. The fundamental crease pattern shown covers 1/16 of the entire crease pattern. The repeated module in the vertical direction folds to an identical or 360° rotated position. This pattern does not rigidly fold.

crease patterns are not rigidly foldable, even if they are triangulated. However, this is not true. Through numerical simulation of different rigid origami structures, we can observe that triangulated periodic crease patterns are generically rigid-foldable, forming a 2-DOF structure, with some special cases forming a 1-DOF mechanism (Section 4).

This mechanical flexibility originates in degeneracy caused by the periodic symmetry of the pattern. Note that Lemma 3.5 converts one identity of rigid body motions along a loop, represented by six equations, into commutativity of two rigid motions. Since the commutativity can be achieved by making the motions share a screw axis, six equations are reduced to four equations. More formally, we can consider a fundamental crease pattern with a triangular hole, whose rigid foldability is same as the rigid foldability of the original mesh (Figure 6). Subsequently, rigid foldability is the intersection of (1) single-vertex compatibility for each interior vertex, and (2) the identity of the relative folding along a curve around the hole, which is converted to the commutativity of two rigid motions, yielding (generically) two degrees of freedom.

4. Numerical Calculation

In this section, we discuss the behaviors of different patterns through numerical kinematics. First, we extract the fundamental crease pattern, so that the boundary does not pass the vertex. The continuous rigid folding motion is simulated by using either a hinge based model or a truss model as the base kinematic simulation, and adding the symmetry constraints that make the corresponding fold angles equal. A

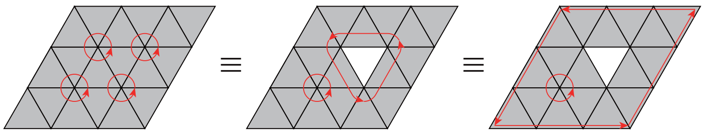

Figure 6. Triangular pattern with hole.

hinge-based model is the system where the configuration is represented by fold angles that are constrained around each vertex, an example of which is *rigid origami simulator* [**5**]. In this system, we get a global folding motion by iteratively integrating a valid infinitesimal motion obtained from Eqution . We employ the truss model, where vertex coordinates are constrained to make the length of the edges constant. We implemented such a system based on the rigid origami simulation mode of Freeform Origami [**6**], while adding the equal angle constraints $\rho_e - \rho_f = 0$ for each symmetric pair e and f. In the case of the truss model, the system forms a system with $n + 6$ variables, and n constraints where $(n = 3V = E)$. Since 6 DOF correspond the rigid body motion of the whole model, they can be eliminated by orienting and fixing one of the facet. In this system, the Jacobian matrix for each infinitesimal step is an $(n+6) \times (n+6)$ square matrix. If there is a continuous rigid folding motion, the matrix is singular with rank $m(< n + 6)$. We obtain a folding motion as a path on an m-dimensional manifold solution space in $n+6$ dimensional configuration space.

4.1. Generic Patterns. Here, we show examples of periodic origami tessellations that exhibit a 2-DOF mechanism. One of the two degrees of freedom in the mechanism represents a "folding-unfolding" motion, and the other represents a "twisting" motion, i.e., a change in the orientation of the axis of cylinder.

4.1.1. *Resch's Triangulated Pattern.* The fundamental figure of Resch's triangulated pattern has 4 distinct vertices and 12 distinct edges. The degeneracy described by Theorem 3.5 gives a continuous folding motion, with two degrees of freedom (Figure 7). Although the crease pattern itself has a three-fold rotational symmetry, the periodic folding does not. As seen in Figure 1, a smaller finite portion is rigid foldable with rotational symmetry, but this is blocked when it becomes big. If we break the rotational symmetry and chose one cylindrical axis, we can continuously fold tessellation of any size to and from two extreme states, i.e., the unfolded and completely folded states (which are rotationally symmetric).

4.1.2. *Waterbomb Tessellations.* Figure 8 shows the behavior of waterbomb tessellations, similarly exhibiting a 2-DOF configuration space.

4.1.3. *Cylindrical Miura-ori.* The pattern shown in Figure 9 is a variation of regular Miura-ori, which folds into a cylindrical surface. If each quadrangular panel is planar, this forms a 1-DOF rigid foldable mechanism. If we split each of the quadrangles into two triangles, we obtain a generic 2-DOF mechanism, allowing for the twist motion that utilize the twist of each triangulated quadrangular panel.

4.2. Degenerate Case: Miura-ori. The pattern shown in Figure 10 is a regular version of Miura-ori, which forms an approximately a planar surface, with a 1-DOF rigid foldable mechanism if each panel is planar. Unlike the cylindrical version, regular version only has 1 DOF, even if triangulated. The pattern can be

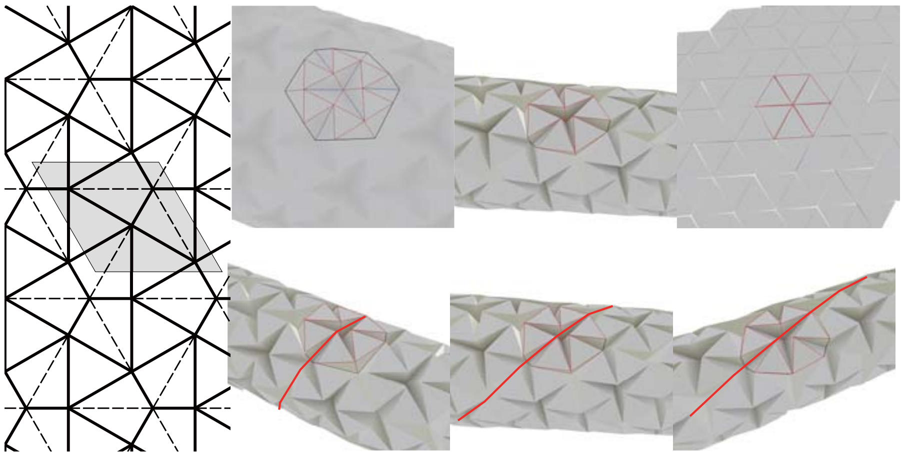

FIGURE 7. Symmetric rigid folding of Ron Resch's triangular tessellation with symmetry constraints. It forms a 2-DOF mechanism; one of the two degrees corresponds to folding-unfolding, and the other to a twisting motion.

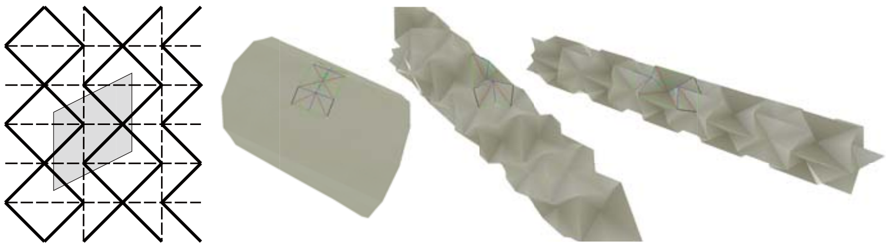

FIGURE 8. Twisting motion in the waterbomb tessellation.

FIGURE 9. Twisting motion of cylindrical Miura-ori.

rigidly folded without folding diagonal creases, but does not allow the possibility to be twisted utilizing the diagonal foldlines. Such a folded form is in a singular

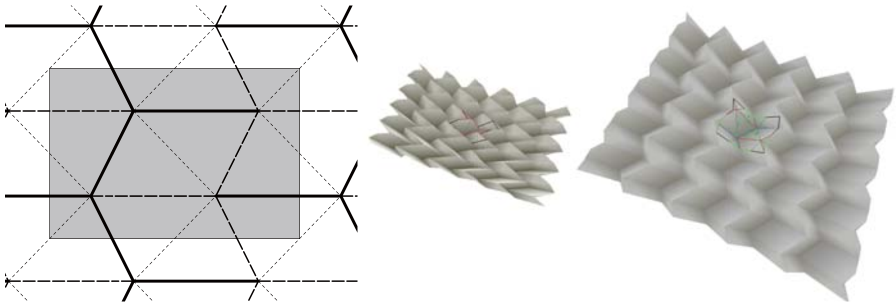

FIGURE 10. Regular Miura-ori cannot be twisted.

configuration. The singularity can be examined by using the infinitesimal folding mode obtained from Equation 4, which can be expressed as a 12×12 matrix. Interestingly, the rank of the matrix is 9, instead of 10, and there exists 3 DOF infinitesimal folding modes; however, two of them do not represent a valid finite folding.

5. Conclusion

We have demonstrated that a generic periodic triangulated origami rigidly folds to a cylindrical form. The resulting structure has 2 DOF; this flexibility can be explained in terms of the periodic symmetry of the pattern. Some exceptions exist; the folded state of an origami torus has no rigid folding motion, and a triangulated regular Miura-ori has only 1 DOF, and thus cannot twist. Rigid foldability supports the manufacturability of these tessellations using different sheet materials. Potential applications include the roll forming of fold core structures, used for sandwich panels. In this paper, we focused only on the wallpaper group $p1$, since it is a subgroup of any other wallpaper group. Further investigation into rigid origami with other symmetry groups would be an interesting direction for future work.

Acknowledgment

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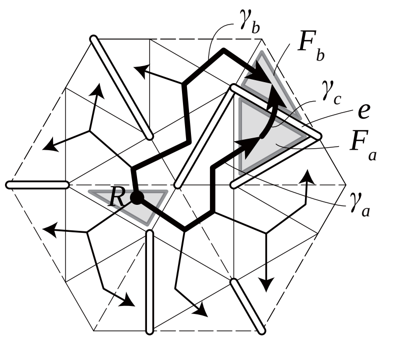

FIGURE 11. Compatibility at edge e.

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Appendix

PROOF OF THEOREM 2.4. The necessity is obvious, i.e., the existence of a rigid folding yields Equation 2.2 for any loop, because otherwise, it is not a valid mapping. In addition, the existence of a rigid folding is equivalent to the existence of rigid folding whose fold angles match the set of fold angles satisfying Equation 2.2.

Sufficiency is shown as follows. Construct a mapping f, by spanning a tree structure starting from a root facet R, so that connecting edges have fold angles satisfying Equation 2.2. Then, construct a path γ for each facet F, such that it starts from the center of R and ends at the center of F and crosses the creases in the sequence of the tree construction. Since each facet is isometrically mapped, this mapping is a valid rigid folding if there is no gap between adjacent facets. Consider a crease e, shared by facets F_a zand F_b, whose paths from R are γ_a and γ_b respectively (Figure 11). Consider a path γ_c, connecting the endpoints of γ_a and γ_b and crossing only crease e. The closed loop consisting of γ_a, γ_c and $\overleftarrow{\gamma}_b$ gives a constraint $\mathbf{F}(\gamma_a)\mathbf{F}(\gamma_c)\mathbf{F}(\gamma_b)^{-1} = \mathbf{I}$. Consider an arbitrary point $\mathbf{x}$ on e; and points $\mathbf{x}_a$ on F_a and $\mathbf{x}_b$ on F_b, both approaching $\mathbf{x}$ from opposite sides. Therefore,

$$\lim_{\mathbf{x}_a \to \mathbf{x}} f(\mathbf{x}_a) = \mathbf{F}(\gamma_a)\mathbf{F}(\gamma_c)\mathbf{x} = \mathbf{F}(\gamma_b)\mathbf{x} = \lim_{\mathbf{x}_b \to \mathbf{x}} f(\mathbf{x}_b). \tag{5.1}$$

We can well define $f(\mathbf{x}) = \lim_{\mathbf{x}_a \to \mathbf{x}} f(\mathbf{x}_a)$, and thus f is continuous at x. Similarly, the mapping is continuous at vertices and thus the entire mapping is a valid folding. □

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